

EDEXCEL INTERNATIONAL A LEVEL

WMA11/01 January 2021 Worked Solutions

Jan 2021 paper rebuilt with the worked solution after each question.

9

paper questions

WMA11

Pure 1

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**

Prepared for Dr Eslam Ahmed - eliteigcse.com

P1

PAST PAPER

WMA11/01 January 2021

Jan 2021 | 9 questions | 75 marks

9

questions

75

marks

Answers

worked solution
after each
question

Question 1

Differentiation

1. A curve has equation

$$y = 2x^3 - 5x^2 - \frac{3}{2x} + 7 \quad x > 0$$

- (a) Find, in simplest form, $\frac{dy}{dx}$ (3)

The point P lies on the curve and has x coordinate $\frac{1}{2}$

- (b) Find an equation of the normal to the curve at P , writing your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found. (5)

(Total 8 marks)

Worked Solution - Question 1

1. Rewrite the negative power term

$$y = 2x^3 - 5x^2 - \frac{3}{2x} + 7 = 2x^3 - 5x^2 - \frac{3}{2}x^{-1} + 7.$$

2. Differentiate term by term

$$\frac{dy}{dx} = 6x^2 - 10x + \frac{3}{2}x^{-2} = 6x^2 - 10x + \frac{3}{2x^2}.$$

3. Find the point P

$$\text{At } x = \frac{1}{2}, y = 2\left(\frac{1}{8}\right) - 5\left(\frac{1}{4}\right) - \frac{3}{1} + 7 = 3, \text{ so } P = \left(\frac{1}{2}, 3\right).$$

4. Find the normal gradient

$$\text{At } x = \frac{1}{2}, \frac{dy}{dx} = 6\left(\frac{1}{4}\right) - 10\left(\frac{1}{2}\right) + \frac{3}{2(1/4)} = \frac{5}{2}. \text{ The normal gradient is therefore } -\frac{2}{5}.$$

5. Write the normal in integer form

$$y - 3 = -\frac{2}{5}\left(x - \frac{1}{2}\right). \text{ Multiplying by 5 gives } 5y - 15 = -2x + 1, \text{ hence } 2x + 5y - 16 = 0.$$

Final answer

$$(a) \frac{dy}{dx} = 6x^2 - 10x + \frac{3}{2x^2}.$$

$$(b) 2x + 5y - 16 = 0.$$

Question 2

Quadratics

2. A tree was planted.

Exactly 3 years after it was planted, the height of the tree was 2 m.

Exactly 5 years after it was planted, the height of the tree was 2.4 m.

Given that the height, H metres, of the tree, t years after it was planted, can be modelled by the equation

$$H^3 = pt^2 + q$$

where p and q are constants,

- (a) find, to 3 significant figures where necessary, the value of p and the value of q .

(4)

Exactly T years after the tree was planted, its height was 5 m.

- (b) Find the value of T according to the model, giving your answer to one decimal place.

(2)

(Total 6 marks)

Worked Solution - Question 2

1. Substitute the two given heights

When $t = 3$ and $H = 2$, $2^3 = 9p + q$, so $8 = 9p + q$. When $t = 5$ and $H = 2.4$, $2.4^3 = 25p + q$, so $13.824 = 25p + q$.

2. Solve for p and q

Subtracting the two equations gives $16p = 5.824$, so $p = 0.364$. Then $8 = 9(0.364) + q$, giving $q = 4.724$, so $q = 4.72$ to 3 significant figures.

3. Use $H = 5$

For $H = 5$, $125 = 0.364T^2 + 4.724$.

4. Find T

$T^2 = \frac{125 - 4.724}{0.364} = 330.428\dots$, so $T = 18.177\dots$. Therefore $T = 18.2$ years to one decimal place.

Final answer

(a) $p = 0.364$, $q = 4.72$.

(b) $T = 18.2$ years.

Question 3

Trigonometric Functions

3.

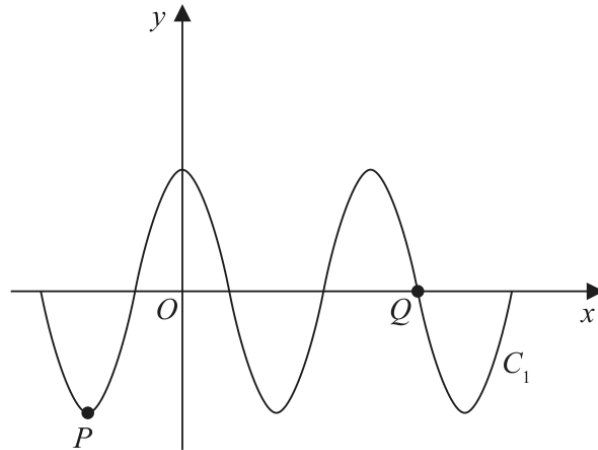


Figure 1

Figure 1 shows a sketch of part of the curve C_1 with equation $y = 4 \cos x^\circ$

The point P and the point Q lie on C_1 and are shown in Figure 1.

(a) State

(i) the coordinates of P ,

(ii) the coordinates of Q .

(3)

The curve C_2 has equation $y = 4 \cos x^\circ + k$, where k is a constant.

Curve C_2 has a minimum y value of -1

The point R is the maximum point on C_2 with the smallest positive x coordinate.

(b) State the coordinates of R .

(2)

(Total 5 marks)

Worked Solution - Question 3

1. Read the key points of $y = 4 \cos x$

For $y = 4 \cos x^\circ$, the maximum value is 4 and the minimum value is -4 . The minimum immediately to the left of the y-axis is at $x = -180^\circ$, so $P = (-180, -4)$.

2. Find Q from the cosine zero

The x-intercepts of $4 \cos x^\circ$ occur at $x = 90^\circ, 270^\circ, 450^\circ, \dots$. The point shown to the right of the next maximum is $Q = (450, 0)$.

3. Use the shifted curve C2

For C_2 , $y = 4 \cos x^\circ + k$. Its minimum is $-4 + k$. Since the minimum value is -1 , $-4 + k = -1$, so $k = 3$.

4. Find R

The maximum value of C_2 is $4 + 3 = 7$. The maximum point with the smallest positive x-coordinate occurs at $x = 360^\circ$, so $R = (360, 7)$.

Final answer

(a)(i) $P = (-180, -4)$. (a)(ii) $Q = (450, 0)$.

(b) $R = (360, 7)$.

Question 4

Straight Line

4.

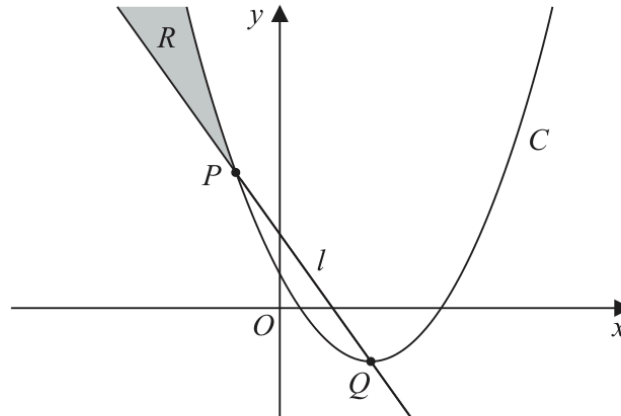


Figure 2

The points P and Q , as shown in Figure 2, have coordinates $(-2, 13)$ and $(4, -5)$ respectively.

The straight line l passes through P and Q .

- (a) Find an equation for l , writing your answer in the form $y = mx + c$, where m and c are integers to be found. (3)

The quadratic curve C passes through P and has a minimum point at Q .

- (b) Find an equation for C . (3)

The region R , shown shaded in Figure 2, lies in the second quadrant and is bounded by C and l only.

- (c) Use inequalities to define region R . (2)

(Total 8 marks)

Worked Solution - Question 4

1. Find the line through P and Q

The gradient is $\frac{-5 - 13}{4 - (-2)} = \frac{-18}{6} = -3$. Using $P(-2, 13)$,

$$y - 13 = -3(x + 2), \text{ hence } y = -3x + 7.$$

2. Use vertex form for the quadratic

The minimum point is $Q(4, -5)$, so the curve has form $y = a(x - 4)^2 - 5$.

3. Substitute P to find a

Since $P(-2, 13)$ lies on the curve, $13 = a(-6)^2 - 5$. Therefore $18 = 36a$ and $a = \frac{1}{2}$.

4. Write the curve

C has equation $y = \frac{1}{2}(x - 4)^2 - 5$.

5. Define the shaded region

In the shaded second-quadrant region, points are above the line, below the curve, and to the left of P . Therefore $y > -3x + 7$, $y < \frac{1}{2}(x - 4)^2 - 5$, and $x < -2$.

Final answer

(a) $y = -3x + 7$.

(b) $y = \frac{1}{2}(x - 4)^2 - 5$.

(c) $y > -3x + 7$, $y < \frac{1}{2}(x - 4)^2 - 5$, $x < -2$.

Question 5

Radians

5.

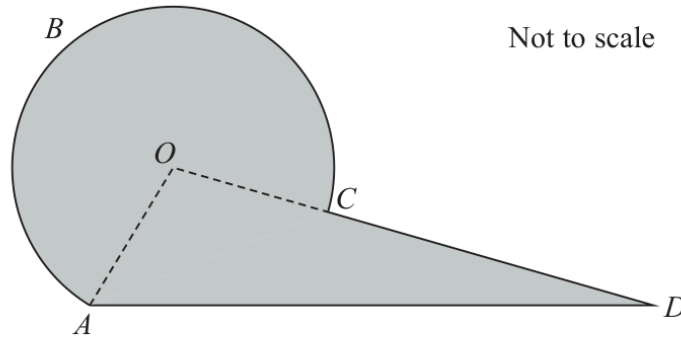


Figure 3

Figure 3 shows the plan view of a viewing platform at a tourist site.

The shape of the viewing platform consists of a sector $ABCOA$ of a circle, centre O , joined to a triangle AOD .

Given that

- $OA = OC = 6\text{ m}$
- $AD = 14\text{ m}$
- angle $ADC = 0.43$ radians
- angle AOD is an obtuse angle
- OCD is a straight line

find

- (a) the size of angle AOD , in radians, to 3 decimal places, (3)
- (b) the length of arc ABC , in metres, to one decimal place, (2)
- (c) the total area of the viewing platform, in m^2 , to one decimal place. (4)

(Total 9 marks)

Worked Solution - Question 5

1. Use the sine rule in triangle AOD

Since O, C, D are in a straight line, $\angle ADO = 0.43$. By the sine rule,

$$\frac{14}{\sin \angle AOD} = \frac{6}{\sin 0.43}, \text{ so } \sin \angle AOD = \frac{14 \sin 0.43}{6}.$$
2. Choose the obtuse angle

The calculator gives the acute value $1.337\dots$, but $\angle AOD$ is obtuse. Hence $\angle AOD = \pi - 1.337\dots = 1.805\dots$, so $\angle AOD = 1.805$ radians.

3. Find the major-sector angle

Arc ABC is the major arc, so its angle is $2\pi - 1.805\dots = 4.478\dots$ radians.

4. Find the arc length

$s = r\theta = 6(4.478\dots) = 26.868\dots$, so the length of arc ABC is **26.9** m.

5. Find the sector area

The sector area is $\frac{1}{2}r^2\theta = \frac{1}{2}(6)^2(4.478\dots) = 80.607\dots$

6. Find the triangle area and total area

$\angle OAD = \pi - 0.43 - 1.805\dots = 0.9066\dots$. The triangle area is $\frac{1}{2}(6)(14)\sin(0.9066\dots) = 33.093\dots$. Total area $= 80.607\dots + 33.093\dots = 113.7 \text{ m}^2$.

Final answer

(a) 1.805 rad.

(b) 26.9 m.

(c) 113.7 m².

Question 6

Graphs of Functions

6. (a) Sketch the curve with equation

$$y = -\frac{k}{x} \quad k > 0 \quad x \neq 0 \quad (2)$$

- (b) On a separate diagram, sketch the curve with equation

$$y = -\frac{k}{x} + k \quad k > 0 \quad x \neq 0$$

stating the coordinates of the point of intersection with the x -axis and, in terms of k , the equation of the horizontal asymptote.

(3)

- (c) Find the range of possible values of k for which the curve with equation

$$y = -\frac{k}{x} + k \quad k > 0 \quad x \neq 0$$

does not touch or intersect the line with equation $y = 3x + 4$

(5)

(Total 10 marks)

Worked Solution - Question 6

1. Sketch $y = -k/x$

Because $k > 0$, $y = -\frac{k}{x}$ has one branch in quadrant II and one branch in quadrant IV, with asymptotes $x = 0$ and $y = 0$.

2. Translate the graph upward

$y = -\frac{k}{x} + k$ is $y = -\frac{k}{x}$ translated up by k units. The vertical asymptote is still $x = 0$, and the horizontal asymptote is $y = k$.

3. Find the x-intercept

Set $y = 0$: $0 = -\frac{k}{x} + k$. Since $k > 0$, $\frac{k}{x} = k$, so $x = 1$. The x-intercept is $(1, 0)$.

4. Set up the intersection equation

For intersections with $y = 3x + 4$, solve $-\frac{k}{x} + k = 3x + 4$. Multiplying by x gives $-k + kx = 3x^2 + 4x$.

5. Use the discriminant

This rearranges to $3x^2 + (4 - k)x + k = 0$. For no touch or intersection, the discriminant must be negative: $(4 - k)^2 - 12k < 0$.

6. Solve the quadratic inequality in k

$k^2 - 20k + 16 < 0$. The roots are $k = 10 \pm 2\sqrt{21}$, so the required range is $10 - 2\sqrt{21} < k < 10 + 2\sqrt{21}$.

Final answer

(b) $(1, 0)$, $y = k$.

(c) $10 - 2\sqrt{21}$

Question 7**Differentiation**

7. In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

$$f(x) = 2x - 3\sqrt{x} - 5 \quad x > 0$$

- (a) Solve the equation

$$f(x) = 9 \quad (4)$$

- (b) Solve the equation

$$f''(x) = 6 \quad (5)$$

(Total 9 marks)

Worked Solution - Question 7

1. Solve $f(x) = 9$ without calculator methods

$$2x - 3\sqrt{x} - 5 = 9, \text{ so } 2x - 3\sqrt{x} - 14 = 0.$$

2. Use $u = \sqrt{x}$

$$\text{Let } u = \sqrt{x}, \text{ where } u > 0. \text{ Then } x = u^2, \text{ so } 2u^2 - 3u - 14 = 0.$$

3. Factorise and reject the invalid root

$$2u^2 - 3u - 14 = (2u - 7)(u + 2). \text{ Since } u > 0, u = \frac{7}{2}, \text{ so } x = u^2 = \frac{49}{4}.$$

4. Differentiate twice

$$f(x) = 2x - 3x^{1/2} - 5, \text{ so } f'(x) = 2 - \frac{3}{2}x^{-1/2} \text{ and } f''(x) = \frac{3}{4}x^{-3/2}.$$

5. Solve $f''(x) = 6$

$$\frac{3}{4}x^{-3/2} = 6, \text{ so } x^{-3/2} = 8. \text{ Therefore } x^{3/2} = \frac{1}{8}.$$

6. Find x

$$\text{Since } x^{3/2} = (\sqrt{x})^3, \text{ we get } \sqrt{x} = \frac{1}{2}, \text{ hence } x = \frac{1}{4}.$$

Final answer

$$(a) x = \frac{49}{4}.$$

$$(b) x = \frac{1}{4}.$$

Question 8

Graphs of Functions

8.

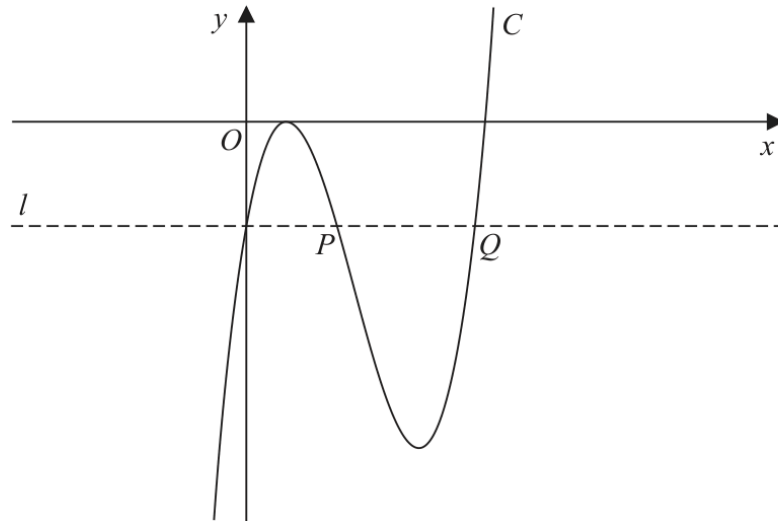


Figure 4

Figure 4 shows a sketch of part of the curve C with equation $y = f(x)$, where

$$f(x) = (3x - 2)^2 (x - 4)$$

(a) Deduce the values of x for which $f(x) > 0$

(1)

(b) Expand $f(x)$ to the form

$$ax^3 + bx^2 + cx + d$$

where a , b , c and d are integers to be found.

(3)

The line l , also shown in Figure 4, passes through the y intercept of C and is parallel to the x -axis.

The line l cuts C again at points P and Q , also shown in Figure 4.

(c) Using algebra and showing your working, find the length of line PQ . Write your answer in the form $k\sqrt{3}$, where k is a constant to be found.

(Solutions relying entirely on calculator technology are not acceptable.)

(5)

(Total 9 marks)

Worked Solution - Question 8

1. Use the factorised form

$f(x) = (3x - 2)^2(x - 4)$. The squared factor is never negative, and it is zero at $x = \frac{2}{3}$. Therefore the sign of $f(x)$ away from the repeated root is controlled by $x - 4$.

2. Find where $f(x)$ is positive

For $f(x) > 0$, we need $x - 4 > 0$, so $x > 4$.

3. Expand $f(x)$

$(3x - 2)^2 = 9x^2 - 12x + 4$. Therefore

$$f(x) = (9x^2 - 12x + 4)(x - 4) = 9x^3 - 48x^2 + 52x - 16.$$

4. Find the line l

The y-intercept of C is $f(0) = -16$, so the horizontal line l is $y = -16$.

5. Find the other intersections with l

Solve $f(x) = -16$: $9x^3 - 48x^2 + 52x - 16 = -16$, so $x(9x^2 - 48x + 52) = 0$.

The root $x = 0$ is the y-intercept, and the two other roots give P and Q .

6. Solve the quadratic

$9x^2 - 48x + 52 = 0$. Hence

$$x = \frac{48 \pm \sqrt{48^2 - 4(9)(52)}}{18} = \frac{48 \pm 12\sqrt{3}}{18} = \frac{8 \pm 2\sqrt{3}}{3}.$$

7. Find PQ

Because P and Q lie on the same horizontal line, PQ is the difference in their x-coordinates:

$$PQ = \frac{8 + 2\sqrt{3}}{3} - \frac{8 - 2\sqrt{3}}{3} = \frac{4\sqrt{3}}{3}.$$

Final answer

(a) $x > 4$.

(b) $f(x) = 9x^3 - 48x^2 + 52x - 16$.

(c) $PQ = \frac{4\sqrt{3}}{3}$.

Question 9

Integration

9. (i) Find

$$\int \frac{(3x + 2)^2}{4\sqrt{x}} dx \quad x > 0$$

giving your answer in simplest form.

(5)

(ii) A curve C has equation $y = f(x)$.

Given

- $f'(x) = x^2 + ax + b$ where a and b are constants
- the y intercept of C is -8
- the point $P(3, -2)$ lies on C
- the gradient of C at P is 2

find, in simplest form, $f(x)$.

(6)

(Total 11 marks)

Worked Solution - Question 9

1. Expand and simplify the integrand

$$\frac{(3x+2)^2}{4\sqrt{x}} = \frac{9x^2 + 12x + 4}{4x^{1/2}} = \frac{9}{4}x^{3/2} + 3x^{1/2} + x^{-1/2}.$$

2. Integrate term by term

$$\int \left(\frac{9}{4}x^{3/2} + 3x^{1/2} + x^{-1/2} \right) dx = \frac{9}{10}x^{5/2} + 2x^{3/2} + 2x^{1/2} + c.$$

3. Write in simplest form

Factoring out $\sqrt{x}$ gives $\frac{\sqrt{x}(9x^2 + 20x + 20)}{10} + c.$

4. Use the gradient condition at P

Since $f'(x) = x^2 + ax + b$ and the gradient at $P(3, -2)$ is 2, $f'(3) = 2$. Hence $9 + 3a + b = 2$, so $3a + b = -7$.

5. Integrate $f'(x)$

$f(x) = \frac{x^3}{3} + \frac{a}{2}x^2 + bx + c$. The y-intercept is -8 , so $c = -8$.

6. Use $P(3, -2)$

$$-2 = \frac{27}{3} + \frac{9a}{2} + 3b - 8, \text{ so } \frac{9a}{2} + 3b = -3.$$

7. Solve for a and b

From $3a + b = -7$, $b = -7 - 3a$. Substitute into $\frac{9a}{2} + 3b = -3$ to get $a = -4$, and then $b = 5$.

8. Write $f(x)$

Therefore $f(x) = \frac{x^3}{3} - 2x^2 + 5x - 8$.

Final answer

$$(i) \frac{\sqrt{x}(9x^2 + 20x + 20)}{10} + c. \quad (ii) f(x) = \frac{x^3}{3} - 2x^2 + 5x - 8.$$